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Things to Check Before Enabling Security Group Writeback to AD

*The setup takes 20 minutes.
The preparation takes 3 weeks.*

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Why Group Provisioning to AD Still Matters

- Many orgs still rely on on-prem AD for resource access — file servers, printers, legacy apps via Kerberos / LDAP.
- Entra ID is the source of truth for identity — but AD still controls access on the ground.
- Cloud Sync group provisioning bridges that gap: groups managed in Entra, enforced in AD.
- Group Writeback v2 in Entra Connect Sync is officially deprecated — Cloud Sync is the only supported path.



Deploy Cloud Sync — Connect Sync Won't Work

- Group Writeback v2 in Entra Connect Sync is **deprecated and no longer supported**.
- You MUST use **Microsoft Entra Cloud Sync (provisioning agent v1.1.1367.0+)**.
- Cloud Sync runs side-by-side with Connect Sync — Cloud Sync handles group provisioning, Connect Sync handles everything else.
- On-prem AD requires **Windows Server 2016+ (msDS-ExternalDirectoryObjectId schema attribute)**.
- Entra Connect Sync **2.2.8.0+** is still needed to sync on-prem users to Entra ID.

No Cloud Sync agent = no security group writeback. Full stop.



Know What's Supported — and What Isn't

SUPPORTED	NOT SUPPORTED
✓ Cloud-created security groups ✓ SOA-converted security groups (migrated from AD) ✓ Assigned membership groups ✓ Dynamic groups (resolved in Entra, written back static)	✗ Microsoft 365 groups (use Group Writeback v1) ✗ Distribution lists — not supported ✗ All written-back groups get Universal scope in AD ✗ Max 50,000 members per group

Map every group you plan to write back. Unsupported types fail silently.



Scope It — Don't Write Back Everything

- Default: you can scope to **ALL security groups** or **SELECTED groups**.
- Scoping to "all" is almost **never what you want in production**.
- Start with a pilot: **select 5–10 specific groups**.
- Use **attribute-based scope filtering** for advanced rules.
- Target container (OU) is set via attribute mapping — use a Switch() expression to route groups to different OUs by displayName.

Unscoped writeback = hundreds of unexpected objects in AD overnight.



Permissions & OU Structure

- The Cloud Sync gMSA service account needs **Read, Write, Create, and Delete** permissions on all descendant groups and users in the target OU.
- Create a **dedicated OU** for written-back groups (e.g., OU=CloudGroups).
- Never write back into production OUs — keep the blast radius contained.
- Permissions are assigned automatically on clean install only — if upgrading, run **Set-AADCloudSyncPermissions** manually.
- These permissions are NOT applied to AdminSDHolder objects by default.
- Verify **AD Recycle Bin** is enabled (your safety net).

If permissions are wrong, writeback fails with no clear error in the portal.



Test On-Demand Before You Enable

- Cloud Sync has **built-in on-demand provisioning** — use it.
- Test with a single group first: validate it appears in the correct OU with the right membership.
- Check that **nested group references resolve correctly** (nested groups must also be in scope).
- Document the current state of target OUs before enabling.
- Know how to disable writeback and remove objects cleanly.
- Configure the **accidental delete protection threshold** before go-live.

The client always asks: "what if we need to undo this?" Have the answer ready.



The Question I Ask Every Client First

*Do you actually need writeback —
or do you need to retire on-prem?*

- Writeback is a bridge, not a destination.
- Every group you write back is a reason to keep AD alive.
- Deploying Cloud Sync just for group provisioning? That's your signal to plan the full migration off Connect Sync.
- The best writeback strategy includes an exit plan.



Before You Enable — Checklist

- ☐ Deploy Cloud Sync provisioning agent (v1.1.1367.0+)
- ☐ Confirm AD DS is Windows Server 2016+ (schema support)
- ☐ Confirm Entra Connect Sync 2.2.8.0+ for on-prem user sync
- ☐ Map supported group types (cloud-created or SOA-converted security groups only)
- ☐ Configure scoping filters (never "all" in production)
- ☐ Set gMSA permissions + dedicated target OU
- ☐ Configure accidental delete threshold
- ☐ Test with on-demand provisioning before full rollout
- ☐ Document rollback procedure

Save this checklist. Run it before every Cloud Sync rollout.

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